FINAL PRESENTATION · 2025

Helpify-AI

Intelligent Customer Support Assistant

NLP + Reinforcement Learning for Intent Classification & Response Optimization — A production-ready system achieving 90.1% accuracy.

NLP · TEXT-CNN | REINFORCEMENT LEARNING | CLINC150 DATASET | 90.1% ACCURACY

PROBLEM STATEMENT

The Problem & Motivation

Why It Matters

- Manual support is **slow, inconsistent, and costly**
- Customers expect **instant, accurate** responses 24/7
- Human agents struggle with **scale & repeatability**

Key Research Questions

- Can we automate **intent detection** reliably?
- Can RL **optimize response decisions**?
- Can we achieve **>90% accuracy** in production?

💡 **Core insight:** Automating support isn't just about speed — it's about making every interaction as good as your best human agent, at infinite scale.

GOALS & RESULTS

Objectives & Success Metrics

Project Goals

- Intent Classification → **≥88%** accuracy
- RL Decision Agent → **≥70%** task success
- Reproducible **one-command pipeline**
- Ethical AI — fair, safe, **privacy-aware**

90.1%
CLASSIFICATION ACCURACY ✅

0.87
MACRO-F1 ✅

74%
RL SUCCESS ✅

DATA

Dataset & Preprocessing

CLINC150 Dataset

- E-commerce domain, curated subset
- **7,998 samples**, 8 intent classes
- **Balanced** class distribution

30,004
VOCAB SIZE

0.8%
OOV RATE

Quality Checks

- ✅ No data leakage
- ✅ Stratified splits
- ✅ Reproducible seed

PROCESSING PIPELINE

Raw Text → Cleaning → Tokenization → Lemmatization → Feature Eng. → Encoding

ARCHITECTURE

Model Architecture
Text-CNN

LAYER STACK

Embedding Layer	GloVe 100d
Parallel CNN Filters	n-gram: 2, 3, 4, 5
Global Max Pooling	Feature extraction
Dense + Dropout	Regularization
Softmax Output	8 intent classes

Why Text-CNN?

- **Fast** ⚡ — ~5ms inference latency
- Captures **n-gram patterns** efficiently
- **Best tradeoff** — accuracy vs. speed
- Scalable to **200 req/sec/core**

> Parallel filters at multiple kernel sizes allow the model to simultaneously detect short phrases and longer expressions.

RL AGENT

Reinforcement Learning Agent

MDP Setup

- **States:** intent + confidence + sentiment = 48 states
- **Actions:** 6 distinct response strategies

⚖️ REWARD DESIGN

Correct + Optimal Response	+2
Correct / Suboptimal	+1
Wrong / Harmful Response	-2

74%

RL TASK SUCCESS RATE ✅

Improvement

Baseline (no RL)	58%
With RL Agent	74%

RL improves decision quality by **+16 percentage points**

RESULTS

Performance Results

90.1%
ACCURACY ↑17.8%

0.87
MACRO F1-SCORE

74%
RL TASK SUCCESS

🏆 Best Intents

shipping_issue	93%
order_status	92%

⚠️ Hardest Intents

refund_request	86%
cancel_order	81%

ANALYSIS

Key Insights — Ablation & Errors

Ablation Study Findings

GloVe Embeddings	+5%
Multi-filter CNN	+3%
RL Agent	+16% task

Error Patterns

- **refund_request** ↔ **complaint** — overlapping semantics
- **order_status** ↔ **shipping_issue** — similar context

🔍 **Main Challenge:** Semantic overlap between related intents. The model confuses intents that share vocabulary but differ in customer goal.

ETHICS & DEPLOY

Ethics, Limitations & Deployment

Ethical Design

- Confidence <70% → escalate to human agent
- Zero personal data storage
- Per-intent fairness monitoring

Limitations

- Single-turn conversations only
- English-only support
- Domain-specific training

Production Metrics

~5ms
INFERENCE LATENCY

200
REQ/SEC PER CORE

CONCLUSION

Takeaways & Future Work

Key Takeaways

- Achieved **production-ready** NLP system
- RL **significantly improves** response decisions (+16%)
- System is **fast, accurate, and scalable**
- Ethical guardrails ensure **safe deployment**

01

Multi-turn

Full conversational context across turns

02

Multilingual

Extend to non-English languages

03

BERT / LLM

Transformer-based model upgrades

04

Real Feedback

Online RL from live user signals

HelpiFy-AI — Intelligent. Scalable. Ethical.

NLP + RL delivering 90.1% accuracy at 5ms latency · Ready for production deployment